Vishal Hegde

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Summary

Software engineer with hands-on experience building enterprise AI platforms, RAG systems, and multi-database backends on Azure at Volvo CE. Strong in Python, Django, PostgreSQL/pgvector, and SQL, with a solid foundation in DBMS, system integration, and secure cloud deployment.

Education

BMS Institute of Technology & Management — Bengaluru, India
B.E. in Electronics & Telecommunication Engineering — *Aug 2022 – May 2026*

- CGPA: 8.13 / 10.00. Coursework: DBMS, DSA, OOP, Operating Systems, Computer Networks, Software Engineering.

Experience

Volvo Construction Equipment — Bengaluru, India
Software Engineering Intern — *Jan 2026 – Present*

- Architected an internal AI marketplace (Django 5.2, Azure App Service, Gunicorn) hosting 15+ GenAI agents for 87 internal Volvo users, with custom Django DB routers isolating each agent into its own PostgreSQL/pgvector database secured via Microsoft Entra ID SSO (MSAL, OAuth2, Volvo-tenant restricted).
- Built two production RAG agents (pgvector, text-embedding-3-large, GPT-4.1) making 86 multi-OEM brochures and 56,853 service records queryable in natural language, with IVFFlat indexing, progressive fallback search, and anti-hallucination grounding.
- Cut LLM cost/latency with a reusable, thread-safe semantic caching library (SHA-256, entity/intent-key, and pgvector-similarity lookup with auto-invalidation), and gave the platform observability via an SDK auto-patch for per-agent cost analytics, a health-check daemon, and a real-time ops dashboard.

Python, Django 5, PostgreSQL/pgvector, Azure App Service, Microsoft Entra ID (MSAL), OpenAI GPT-4.1, text-embedding-3-large, RAG.

Projects

Ark — Deduplicating Backup & Restore Engine | *Python*

- Built a snapshot-based backup engine using content-defined chunking (FastCDC) and BLAKE2b content-addressing, so identical/near-identical data is stored once and incremental backups are near-zero-cost; adaptive compression, AES-256-GCM encryption (scrypt KDF), full CLI, end-to-end tests.

BTreeDB — Disk-Based Embedded Database Engine | *Python*

- Built an on-disk B+-tree engine from scratch (paged storage, write-ahead log for crash recovery, secondary indexes) with a small SQL layer, CLI, and GUI for querying/inspecting the database.

RaftKV — Distributed Key-Value Store on Raft Consensus | *Python*

- Implemented Raft consensus (leader election, log replication, snapshotting) from scratch for a fault-tolerant, strongly-consistent distributed key-value store, with a simulator for testing partitions/failover.

Sentinel — Static Security Scanner for Python, Django & AI-Agent Apps | *Python*

- Built a self-contained SAST tool (AST-based) flagging 20+ vulnerability classes (injection, hardcoded secrets, weak crypto, OWASP LLM Top 10 issues) with taint analysis, CI baseline diffing, and JSON/PDF reports.

Factory Workforce Analytics System | *Python, PostgreSQL, OpenCV, YOLOv8* — Sep 2023 – Dec 2023

- Built a real-time video pipeline for entry/exit tracking and attendance, storing results in PostgreSQL for compliance reporting; research published at IEEE CICCS 2025.

Technical Skills

Languages & Data: Python, SQL (PostgreSQL, MySQL), Java; PostgreSQL/pgvector, Redis; schema design, indexing, query optimization
AI & Backend: RAG, OpenAI GPT-4.1, text-embedding-3-large, semantic search; Django, Django REST Framework, Docker, Git, Gunicorn
Cloud & Auth: Azure App Service, Azure OpenAI, Microsoft Entra ID (MSAL / OAuth2 SSO), AWS (EC2, S3, IAM), Oracle Cloud (OCI)

Research & Publications

- **Workforce Efficiency in Factories: A Computer Vision Approach** – Published, IEEE CICCS 2025 (ieeexplore.ieee.org/document/11280292).
- **AI-Driven Workforce Monitoring & Safety Compliance in Factories Using Computer Vision and the Django Framework** – Published, IEEE COMP-SIF 2026 (ieeexplore.ieee.org/document/11481888).